

Physics and Astronomy Department
The University of North Carolina at Greensboro
1400 Spring Garden Street
Greensboro, North Carolina 27412

February 20, 2026

Applying for the Position of Temporary Instructor of Physics

Dr. Pratap,

I write regarding the exciting opportunity to work as a temporary instructor of Physics I and II at UNC-Greensboro, as advertised via announcement. I am currently a doctoral candidate at UNC-Chapel Hill; I have earned my Master's of Science in Physics and have completed all requirements for my doctorate except my dissertation, which I anticipate defending in Spring 2027. As a Royster Doctoral Fellow with non-service funding, I am available to fill the listed position this spring.

My extensive experience as a physics instructor includes a long history of working with students enrolled in introductory physics courses; I have acted as a teacher's assistant at both Wake Forest and UNC-Chapel Hill for these courses. My time teaching was formative, orienting me towards the efficacy of an experiential learning process where students develop an enhanced understanding of curricula via interactive modules. Additionally, I have eight years as a private physics tutor, which has provided me with first-hand insight into the diverse learning styles of students from all walks of life.

I believe that it is crucial that students see the connection between their curricula and how that knowledge can be applied to what is most important in life. As an educator, I benefit from years of philosophical study, which helps give me a deeply grounded, functionalist approach to physics focused on how theory can be applied dynamically to understand and positively shape the world around us. My educational experiences are also buttressed by years of experimental biophysics research, which gives a broad applied knowledge base, empowering me to teach with expertise and nuance. Recently, I mentored a team of undergraduates, guiding them to build a 3D-printed fluorescent microscope capable of advanced particle tracking from inexpensive, commercially available parts. The team went on to win awards for their work at consecutive annual UNC MakerFest competitions; we are now cooperating with low-income highschools in Eastern NC to conduct these builds locally, giving them access to microscopy tools which would normally cost tens of thousands of dollars. These experiences have prepared me to help students understand that studying physics is about more than lab experiments and a curricular requirement, instead offering an empowering understanding of daily life.

Thank you for your time, and I look forward to talking with you about this exciting opportunity in the coming weeks.

Sincerely,

Alexander Marshall
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Alexander Marshall | Physicist

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Royster Fellow | Doctoral Candidate | UNC-Chapel Hill

Education

Doctorate of Philosophy in Physics

Chapel Hill, NC

The University of North Carolina at Chapel Hill

2027

Royster Doctoral Fellow | Exams Passed | All But Dissertation

Master of Science in Physics

Chapel Hill, NC

The University of North Carolina at Chapel Hill

2025

Completed *en route* to PhD

B.S. in Physics, B.A. in Philosophy, Minor in Math.

Winston-Salem, NC

Wake Forest University

2022

GPA 3.91 | *Summa Cum Laude* | Stamps Scholar

Academic Projects

Doctoral Thesis.....

Anomalous Diffusion in Poroelastic Media

Advisor: Richard Superfine

Thesis Committee: Dr. Richard Superfine, Dr. C.A. Forrest, Dr. David Hill, Dr. Amy Oldenburg, Dr. Ehssan Nazockdast

Description: Explore the role of static and dynamic strains in the transport of large solutes in various poroelastic media, including transcription factor search patterns in cellular nuclei exposed to dynamic strains.

Masters Thesis.....

Physiological Strains in Poroelastic Media Induce Asymmetric Transport.

Advisor: Richard Superfine

Thesis Committee: Dr. Richard Superfine, Dr. Greg M. Forrest, Dr. David Hill, Dr. Amy Oldenburg, Dr. Kerry Bloom

Description: Tracked quantum dots undergoing dynamic diffusion in poroelastic media using LSM with unprecedented temporal resolution. Live imaging during physiological strain reveals that large solute transport is asymmetric during dynamic strains, with advection yielding significantly more forward angular anisotropy during the relaxation stroke than the compression stroke. These results suggest single-particle effects could explain the macroscopic solute concentration enhancements reported by studies of poroelastic media under dynamic strain.

Undergraduate Honors Thesis.....

Generating a 3-D Lattice of Photonic Spheres to Track Fluorescently Labelled Chromatin

Advisor: Dr. Keith Bonin

Committee: Dr. Keith Bonin, Dr. George Holzwarth, Dr. Stephen Baker

Description: Wrote a custom iterative routine to generate, test, and analyze hundreds of thousands of phase settings on a spatial light modulator, implementing the correction method of Dholakia et al. (2017) Used this routine to create 3D lattice of photonic spheres to be used for tracking labelled chromatin in irradiated cells.

Additional Research.....

Synthesizing Chalcogenide Crystals for Quantum Computing **Winston-Salem, NC**

Wake Forest Nanotech Center, Dr. David Carroll 2021

Assisted with the synthesis of chalcogenide crystals being investigated as a potential substrate for quantum computing.

Neurochemical Addiction Mechanisms In the Nocus Accumbens **Winston-Salem, NC**

Wake Forest School of Medicine, Dr. Evgeny Budygin 2019-2021

Conducted cyclic voltammetry on rats *in vivo* and mouse brain slices *in situ* to explore the role of neurotransmitters in addictive behaviors.

Cultural Impact of Oenology in the Peleponnese **Winston-Salem, NC**

Wake Forest Department of Anthropology, Dr. Karen Friederic 2019

Conducted a month-long anthropological study in Nemea, Greece on how the economic significance of the viticultural industry impacts local culture. I found that extreme seasonal workloads intersect with traditional gendered domestic labor expectations to inhibit female career growth in this industry.

Cancer and Postoperative Pain **Winston-Salem, NC**

Wake Forest Baptist Medical Center, Dr. Christopher Peters 2018

Prepared solutions, assisted with operations on research subjects, and conducted fluorescence microscopy on brain slices from rats to ascertain the distribution of optogenetic proteins in neurons after treatment with a viral vector.

Scholarships and Grants

2022-2027: Royster Doctoral Fellowship. Five-year fellowship awarded for academic merit.

2018-2022: Stamps Scholarship. Four-year scholarship awarded for academic merit.

2019: Wake Forest Scholars Travel Grant. Funded summer anthropology project abroad.

Publications

Prestigious Journal A **Chapel Hill, NC**

Strain changes particle behavior in poroelastic media, UNC Chapel Hill 2027

A Marshall, R. Superfine

Prestigious Journal B **Chapel Hill, NC**

Dynamic Strains Drive Asymmetric Flow, UNC Chapel Hill 2026

A. Marshall, R. Superfine

Prestigious Journal C **Chapel Hill, NC**

Histones exhibit confined behavior during interphase, UNC Chapel Hill 2026

A. Marshall, R. Superfine, K. Bloom, Zhang, J

Technical skills

Coding Languages: MATLAB, LabVIEW, Python, L^AT_EX, Git

Scripting: μ Manager, ImageJ

Data Analysis: Particle Tracking, Image Analysis, Stochastic Processes

Other: Transmission Microscopy, Fluorescent Microscopy, Light Sheet Fluorescent Microscopy, Total Internal Reflection Microscopy, Custom Optical Design, Laser Systems, Spatial Light Modulation, Remote Focusing, Machine Vision, Misc. Lab Skills, Hazardous Waste Disposal

Presentations and Symposia

Physiological Strains in Poroelastic Media Induce Asymmetric Transport. Chapel Hill, NC
Master's Defense, UNC-Chapel Hill November 17, 2025

Strain Alters Anomalous Diffusion in Poroelastic Media. Elon, NC
Symposium, Elon University October 1, 2025

Diffusion in Strained Poroelastic Media Chapel Hill, NC
Thesis Proposal, UNC-Chapel Hill September 16, 2025

Generating a Three-Dimensional Lattice of Photonic Spheres Winston-Salem, NC
Symposium, Wake Forest University May 2022

Conferences

Undergraduate Research Day Winston-Salem, NC
Symposium, Wake Forest University URECA Center September 2019

Probing Structural and Force Dynamics during Phagocytosis Raleigh, NC
Triangle Cytoskeleton Meeting, NC Museum of Natural Sciences September 25, 2023

Strain Driven Transport In Poroelastic Media Raleigh, NC
Triangle Cytoskeleton Meeting, NC Museum of Natural Sciences September 23, 2024

Various topics Chapel Hill, NC
Annual Colloquium, UNC Applied Physical Sciences Department Annually, 2022-2025

Strain-Driven Transport In Poroelastic Media Raleigh, NC
8th International Soft Matter Conference, Soft Matter Association of the Americas August 2024

Strain-Driven Transport In Poroelastic Media Durham, NC
Triangle Soft Matter Workshop, Duke University Dept. of Physics May 12, 2025

Strain-Driven Transport In Poroelastic Media Chapel Hill, NC
13th Carolina Biophysics Symposium, UNC-Chapel Hill November 6, 2025

Strain-Driven Transport In Poroelastic Media Chapel Hill, NC
Royster Global Conference, Royster Society of Fellows October 21, 2025

Strain Driven Transport In Poroelastic Media Durham, NC
Triangle Cytoskeleton Meeting, Duke University September 15, 2025

Teaching and Mentorship

Undergraduate Research Mentorship

Chapel Hill, NC

Anomalous Diffusion in Strained Poroelastic Media

2023-Present

Undergraduate involvement is integral to the research for my doctoral dissertation. I have collaborated with up to six concurrent students. Many of these relationships have developed into multi-year collaborations which contribute massively to the project. Over the course of their involvement, students have learned cutting-edge techniques and developed as scholars; several now conduct related projects largely independently.

Teaching Assistant

Winston-Salem, NC

Physics I, Wake Forest University

Fall 2021

Due to COVID-19 restrictions, the instructor of record was unable to acquire a work visa and could not enter the United States. In her absence, I took on additional responsibilities for the course.

Teaching Assistant

Chapel Hill, NC

Physics I, UNC-Chapel Hill

2023-2024

Primarily acted as a lab instructor. Responsible for day-to-day administration of two 110 minute 75 student lab sections which met biweekly. Day-to-day administration of course conducted by TAs. I taught this course three times.

Teaching Assistant

Chapel Hill, NC

Other Teaching Duties, UNC-Chapel Hill

2023-2024

Served as a TA for various other courses. Topics covered included Thermodynamics, Quantum Physics II, a first-year seminar centered on the philosophy of science, working in the departmental tutoring center, and more.

Departmental tutor

Chapel Hill, NC

Physics I and II, UNC Naval Armory

Fall 2023-Spring 2024

Provided physics tutoring for ROTC-enrolled students as part of a departmental initiative.

Private Tutor

Various Locations

STEM-related Topics

2018-Present

Worked one-on-one with students of high-school to college age on topics ranging from high school chemistry to Physics Olympiad preparation.

2022-Present: Undergraduate Research Mentorship

Mentored many undergraduate students.

Outreach

Open Flexure Microscope and Molecular Stamper

Chapel Hill, NC

UNC-Chapel Hill

2023-Present

Mentored a team of undergraduate researchers as they built OpenFlexure microscopes, ultra-low cost devices manufactured with 3-D printed parts which nevertheless possess advanced functionality. Undergraduates won prizes at annual UNC MakerFest competition for two consecutive years. Initiatives planned for building microscopes in low-resource highschools in Eastern NC.

North Carolina Science Festival

Chapel Hill, NC

Morehead Planetarium and Science Center

2023-Present

Each year, during this public outreach event the Superfine research group conducts science demos and research talks targeted at general audiences.

Hospital Volunteer <i>Wake Forest Baptist Medical Center</i> Volunteered in various patient-facing positions, accumulating over 300 service hours.	Winston-Salem, NC <i>2015–2017</i>
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References

Richard Superfine, PhD <i>superfine@unc.edu, (919) 432-4381</i> Taylor-Williams Distinguished Professor, UNC-Chapel Hill	Doctoral Advisor <i>2022–Present</i>
Kerry Bloom, PhD <i>kerry_bloom@unc.edu, (919) 962-1182</i> Thad L. Beyle Distinguished Professor, UNC-Chapel Hill	Reasearch Collaborator <i>2025–Present</i>
David Hill, PhD <i>dbhill@email.unc.edu, (919) 962-4718</i> Associate Professor, UNC-Chapel Hill	Doctoral Committee Member <i>2025–Present</i>
Keith Bonin, PhD <i>bonin@wfu.edu, (336) 758-4962</i> Vice Provost, Wake Forest University	Undergraduate Advisor <i>2019–2022</i>
Martin Guthold, PhD <i>guthold@wfu.edu, (336) 758-4977</i> Physics Dept. Chair, Wake Forest University	Personal Reference <i>2010–Present</i>

Languages

English: Fluent	<i>Native Speaker</i>
Modern Greek: Intermediate	<i>Conversational speaker.</i>
Classical Latin: Intermediate	<i>Seven years of formal instruction</i>